Square-Summable Sequences and Sequences Without Convergent Subsequences

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Source: MIT OCW 18.100B Real Analysis (Spring 2025), Problem Set 7, Problems 1 and 2

Problem 1: The Square-Summable Sequence Space

Problem Statement. Let

$$X = \Big\{(a_1, a_2, a_3, \dots) : \sum_{j=1}^{\infty} a_j^2 \text{ converges}\Big\},$$

and for $x = (a_1, a_2, \dots)$ and $y = (b_1, b_2, \dots)$ in X, define

$$d(x, y) = \sqrt{\sum_{j=1}^{\infty} (a_j - b_j)^2}.$$

(1) Show that d is well-defined, i.e. that $\sum_{j=1}^{\infty}(a_j - b_j)^2$ converges for all $x, y \in X$.

(2) Show that d satisfies the triangle inequality. (The triangle inequality in $\mathbb{R}^n$ may be used without proof.)

(3) Let $x_n = (0, \dots, 0, 1, 0, \dots)$, with 1 in the n-th position. Show that (x_n) has no convergent subsequence.

Part (1)

Proof. Let $x = (a_j)$ and $y = (b_j)$ be in X. For any real numbers s and t,

$$(s - t)^2 \le (s - t)^2 + (s + t)^2 = 2s^2 + 2t^2.$$

Applying this with $s = a_j$ and $t = b_j$ gives

$$0 \le (a_j - b_j)^2 \le 2a_j^2 + 2b_j^2 \qquad \text{for every } j.$$

Since $\sum a_j^2$ and $\sum b_j^2$ converge, so does $\sum(2a_j^2 + 2b_j^2)$, by the algebraic limit theorem for series. By the comparison test, $\sum_{j=1}^{\infty}(a_j - b_j)^2$ converges. Its sum is nonnegative, so its square root $d(x, y)$ is a well-defined real number. □

Part (2)

Proof. Let $x = (a_j)$, $y = (b_j)$, and $z = (c_j)$ be in X. We must show $d(x, z) \le d(x, y) + d(y, z)$.

Fix $n \ge 1$. By the triangle inequality in $\mathbb{R}^n$,

$$\sqrt{\sum_{j=1}^{n}(a_j - c_j)^2} \le \sqrt{\sum_{j=1}^{n}(a_j - b_j)^2} + \sqrt{\sum_{j=1}^{n}(b_j - c_j)^2}.$$

The series $\sum(a_j - b_j)^2$ has nonnegative terms, so each partial sum is at most the full sum, which exists by Part (1). Since the square root is increasing,

$$\sqrt{\sum_{j=1}^{n}(a_j - b_j)^2} \le d(x, y), \qquad \text{and likewise} \qquad \sqrt{\sum_{j=1}^{n}(b_j - c_j)^2} \le d(y, z).$$

Combining these,

$$\sqrt{\sum_{j=1}^{n}(a_j - c_j)^2} \le d(x, y) + d(y, z) \qquad \text{for every } n.$$

As $n \to \infty$, the partial sums $\sum_{j=1}^{n}(a_j - c_j)^2$ converge to $\sum_{j=1}^{\infty}(a_j - c_j)^2$ by Part (1). Since the square root is continuous, the left-hand side converges to $d(x, z)$. Non-strict inequalities are preserved under limits, so

$$d(x, z) \le d(x, y) + d(y, z).$$

□

Part (3)

Proof. First, each x_n lies in X, since the sum of the squares of its entries is 1.

Distance between distinct terms. Let $n \ne m$. The sequences x_n and x_m differ in exactly two positions, the n-th and the m-th, and the difference in each is ± 1. Hence

$$d(x_n, x_m) = \sqrt{1^2 + 1^2} = \sqrt{2}.$$

No subsequence converges. Let (x_{n_k}) be any subsequence, where $n_1 < n_2 < n_3 < \cdots$. Suppose, for the sake of contradiction, that $x_{n_k} \to y$ for some $y \in X$. Taking $\varepsilon = \frac{\sqrt{2}}{2}$, there exists K such that $d(x_{n_k}, y) < \frac{\sqrt{2}}{2}$ for all $k \ge K$. Since $n_K \ne n_{K+1}$, the triangle inequality from Part (2) and the symmetry of d give

$$\sqrt{2} = d(x_{n_K}, x_{n_{K+1}}) \le d(x_{n_K}, y) + d(y, x_{n_{K+1}}) < \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2} = \sqrt{2},$$

which is a contradiction. Therefore (x_n) has no convergent subsequence. □

Problem 2: Every Term Is Isolated

Problem Statement. Let (X, d) be a metric space and let (x_n) be a sequence in X with no convergent subsequence. Let $E = \{x_1, x_2, x_3, \dots\}$. Show that for every $k \in \mathbb{N}$ there exists $r_k > 0$ such that

$$B(x_k, r_k) \cap E = \{x_k\}.$$

We may use the following without proof.

Given Fact. Let $x \in X$. If $E \cap B(x, r)$ contains infinitely many elements for every $r > 0$, then (x_n) has a subsequence converging to x.

Solution

Proof. Fix $k \in \mathbb{N}$. Note that $x_k \in B(x_k, r) \cap E$ for every $r > 0$, so the set $B(x_k, r) \cap E$ always contains x_k. Thus the statement to prove is that for some $r > 0$, this set contains *no other* point of E.

Suppose, for the sake of contradiction, that no such r exists. Then

$$\text{for every } r > 0, \quad B(x_k, r) \cap E \text{ contains a point other than } x_k. \tag{$*$}$$

We show that this forces $B(x_k, r) \cap E$ to be infinite for every $r > 0$.

Every ball around x_k meets E in infinitely many points. Suppose instead that $B(x_k, r_0) \cap E$ is finite for some $r_0 > 0$. By $(*)$, this set contains at least one point other than x_k. Let $y_1, \ldots, y_m$ be the points of $B(x_k, r_0) \cap E$ that are different from x_k, where $m \geq 1$. Each $y_i \neq x_k$, so $d(x_k, y_i) > 0$. Define

$$r' = \min\{d(x_k, y_1), \ldots, d(x_k, y_m)\}.$$

As the minimum of finitely many positive numbers, $r' > 0$. Moreover, $r' < r_0$, because each $y_i \in B(x_k, r_0)$.

Now consider $B(x_k, r') \cap E$. Since $r' < r_0$, this set is contained in $B(x_k, r_0) \cap E = \{x_k, y_1, \ldots, y_m\}$. However, no y_i lies in $B(x_k, r')$, because $d(x_k, y_i) \geq r'$. Hence

$$B(x_k, r') \cap E = \{x_k\},$$

which contradicts $(*)$ with $r = r'$. Therefore $B(x_k, r) \cap E$ is infinite for every $r > 0$.

Conclusion. By the Given Fact applied to $x = x_k$, the sequence (x_n) has a subsequence converging to x_k. This contradicts the hypothesis that (x_n) has no convergent subsequence.

Hence our assumption was false, and there exists $r_k > 0$ with $B(x_k, r_k) \cap E = \{x_k\}$. Since k was arbitrary, this holds for every $k \in \mathbb{N}$. $\square$